

**TITLE:**

**Cross-Dataset Comparison Report**

**Dataset Used:**

**Adult Psychiatric Morbidity Survey (APMS) 2014**

**&**

**Crime Survey for England & Wales (CSEW) — Mental Health  
Module**

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**ZENO TALENT**

# Cross-Dataset Comparison Report

## APMS vs CSEW Mental Health Analysis

### 1. Dataset Introduction

#### APMS Dataset

The APMS (Adult Psychiatric Morbidity Survey) dataset contains information related to mental health conditions such as anxiety, depression, demographic details, and psychological disorder indicators. It helps analyse mental health prevalence among different population groups.

Main features:

- Anxiety and depression indicators
- Demographic information
- Mental health status classification
- Statistical mental health measures

#### CSEW Dataset

The CSEW (Crime Survey for England and Wales) dataset provides yearly mental health indicators collected from survey respondents. It helps study changes in mental health trends over time, especially before, during, and after the COVID-19 period.

Main features:

- Year-wise mental health indicators
- COVID-era mental health impact
- Population survey responses
- Trend-based mental health statistics

#### Purpose of Cross-Dataset Comparison

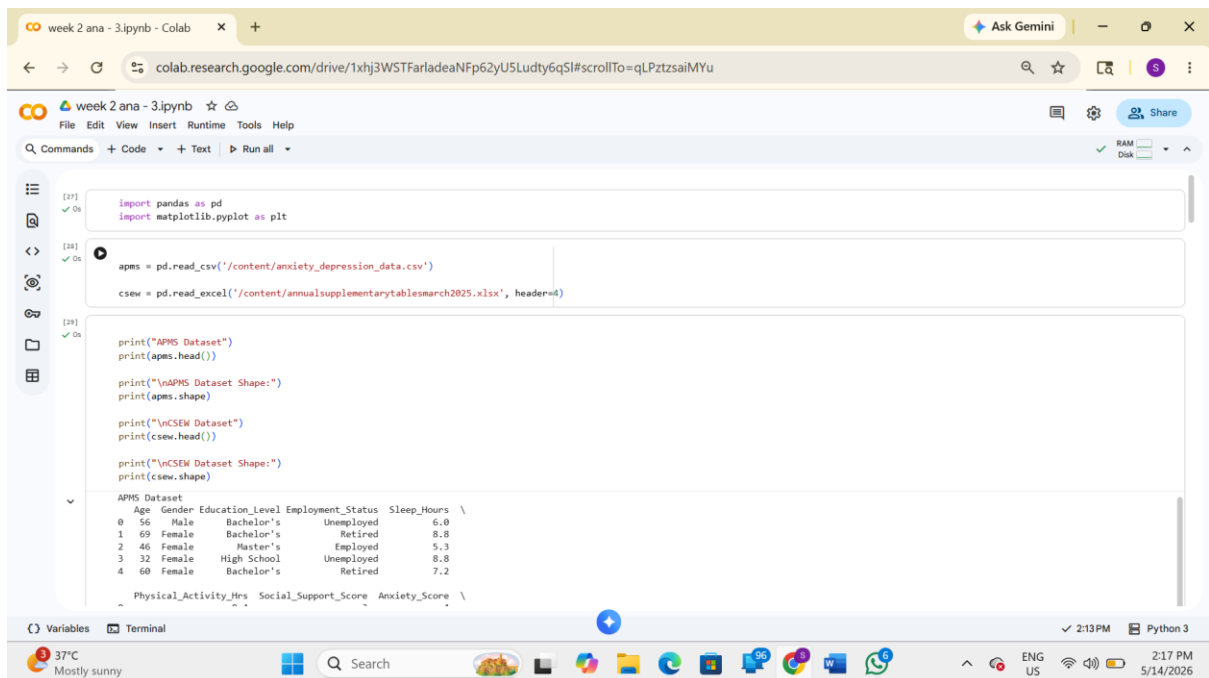
The objective of this analysis is to:

- Compare mental health trends across both datasets
- Identify similarities and differences
- Align shared dimensions such as anxiety and depression indicators
- Detect agreements and conflicts between findings
- Understand how mental health patterns changed over time

# Technologies Used

| Technology            | Purpose                         |
|-----------------------|---------------------------------|
| Python                | Data analysis and processing    |
| Pandas                | Data cleaning and manipulation  |
| Matplotlib            | Data visualization and charts   |
| Google Colab Notebook | Running and sharing the project |

## Cross Dataset Comparison



The screenshot shows a Google Colab notebook interface. The code is as follows:

```
[27] import pandas as pd
import matplotlib.pyplot as plt

[28] apms = pd.read_csv('/content/anxiety_depression_data.csv')
csew = pd.read_excel('/content/annualsupplementarytablesarch2025.xlsx', header=4)

[29] print("APMS Dataset")
print(apms.head())

print("\nAPMS Dataset Shape:")
print(apms.shape)

print("\nCSEW Dataset")
print(csew.head())

print("\nCSEW Dataset Shape:")
print(csew.shape)
```

The output of the code shows the first five rows of the 'APMS Dataset' and its shape (5 rows, 5 columns). The columns are: Age, Gender, Education\_Level, Employment\_Status, and Sleep\_Hours.

|   | Age | Gender | Education_Level | Employment_Status | Sleep_Hours |
|---|-----|--------|-----------------|-------------------|-------------|
| 0 | 56  | Male   | Bachelor's      | Unemployed        | 6.0         |
| 1 | 69  | Female | Bachelor's      | Retired           | 8.8         |
| 2 | 46  | Female | Master's        | Employed          | 5.3         |
| 3 | 32  | Female | High School     | Unemployed        | 8.8         |
| 4 | 60  | Female | Bachelor's      | Retired           | 7.2         |

The bottom of the notebook shows the 'Variables' tab with the following variables: Physical\_Activity\_Hrs, Social\_Support\_Score, and Anxiety\_Score.

Colab interface showing a Jupyter Notebook with the following code and output:

```
Physical_Activity_Hrs Social_Support_Score Anxiety_Score \
0 0.4 3 4
1 2.8 6 18
2 1.6 5 5
3 0.5 4 6
4 0.7 2 7

Depression_Score Stress_Level ... Chronic_Illnesses Medication_Use \
0 2 9 ... 0 NaN
1 7 6 ... 0 NaN
2 13 8 ... 0 NaN
3 3 4 ... 1 NaN
4 15 3 ... 0 NaN

Therapy_Meditation Substance_Use Financial_Stress Work_Stress \
0 0 1 NaN 4 3
1 1 0 NaN 1 4
2 0 1 NaN 8 7
3 0 0 NaN 7 4
4 1 1 Frequent 8 9

Self_Esteem_Score Life_Satisfaction_Score Loneliness_Score
0 7 5 1
1 7 4 6
2 8 1 1
3 8 4 4
4 5 7 7

[5 rows x 21 columns]

APHS Dataset Shape:
(1200, 21)

CSEW Dataset
Data tables shown in this workbook relate to the Crime Survey for England and Wales (CSEW). \
```

Variables Terminal 2:13 PM Python 3

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Colab interface showing a Jupyter Notebook with the following code and output:

```
4 1 1 Frequent 8 9

Self_Esteem_Score Life_Satisfaction_Score Loneliness_Score
0 7 5 1
1 7 4 6
2 8 1 1
3 8 4 4
4 5 7 7

[5 rows x 21 columns]

APHS Dataset Shape:
(1200, 21)

CSEW Dataset
Data tables shown in this workbook relate to the Crime Survey for England and Wales (CSEW). \
0 Crime Survey for England and Wales (CSEW) esti...
1 Due to the suspension of the CSEW during the C...
2 CSEW headline crime captures a range of person...
3 For explanatory notes on these statistics see ...
4 For further information about the Crime Survey...

Unnamed: 1
0 Quality of estimates from the Crime Survey for...
1 NaN
2 NaN
3 User Guide to Crime Statistics for England and...
4 NaN

CSEW Dataset Shape:
(8, 2)

print("APHS Missing Values")
```

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Colab interface showing code execution for missing values in APMS and CSEW datasets.

```
[39] ✓ On
print("APMS Missing Values")
print(apms.isnull().sum())

print("\nCSEW Missing Values")
print(csew.isnull().sum())
```

APMS Missing Values

|                               |       |
|-------------------------------|-------|
| Age                           | 0     |
| Gender                        | 0     |
| Education_Level               | 0     |
| Employment_Status             | 0     |
| Sleep_Hours                   | 0     |
| Physical_Activity_Hrs         | 0     |
| Social_Support_Score          | 0     |
| Anxiety_Score                 | 0     |
| Depression_Score              | 0     |
| Stress_Level                  | 0     |
| Family_History_Mental_Illness | 0     |
| Chronic_Illnesses             | 0     |
| Medication_Use                | 747   |
| Therapy                       | 0     |
| Meditation                    | 0     |
| Substance_Use                 | 834   |
| Financial_Stress              | 0     |
| Work_Stress                   | 0     |
| Self_Esteem_Score             | 0     |
| Life_Satisfaction_Score       | 0     |
| Loneliness_Score              | 0     |
| dtype:                        | int64 |

CSEW Missing Values

Data tables shown in this workbook relate to the Crime Survey for England and Wales (CSEW).

|            |       |
|------------|-------|
| Unnamed: 1 | 0     |
| dtype:     | int64 |

Colab interface showing code execution for plotting the distribution of Anxiety Scores.

```
[10] ✓ On
import pandas as pd
import matplotlib.pyplot as plt

apms_counts = apms['Anxiety_Score'].value_counts().sort_index()

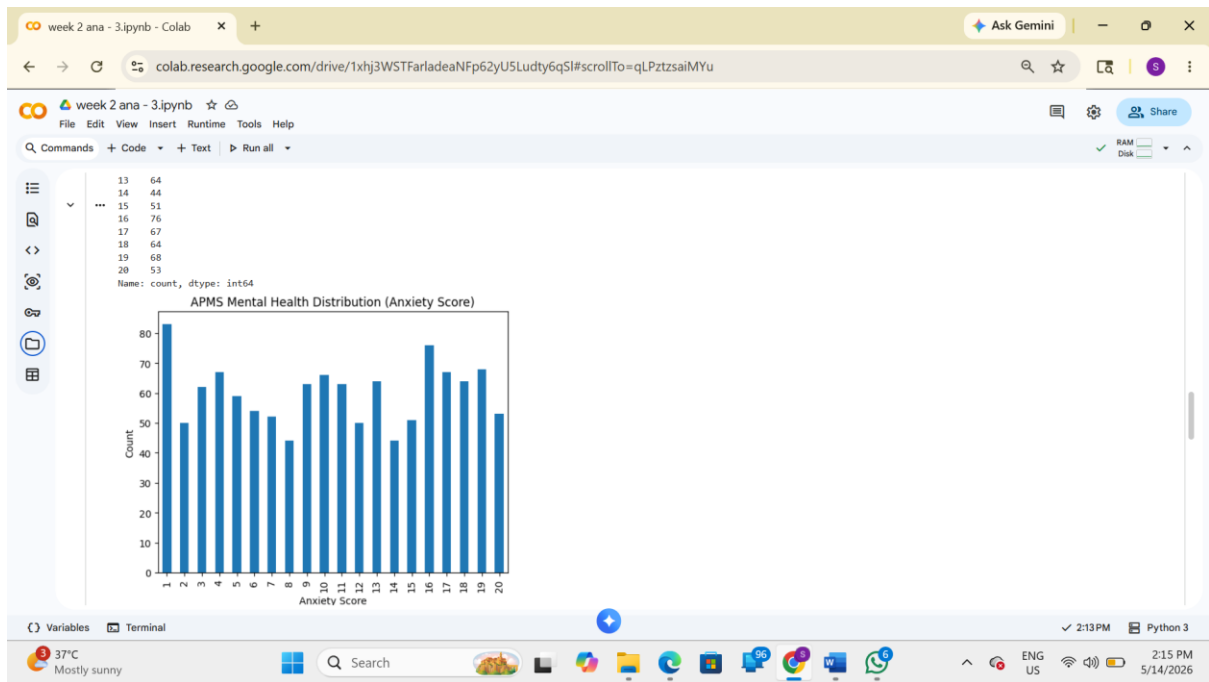
print(apms_counts)

apms_counts.plot(kind='bar')

plt.title('APMS Mental Health Distribution (Anxiety Score)')
plt.xlabel('Anxiety Score')
plt.ylabel('Count')
plt.show()
```

Anxiety\_Score

|    |    |
|----|----|
| 1  | 83 |
| 2  | 50 |
| 3  | 62 |
| 4  | 67 |
| 5  | 59 |
| 6  | 54 |
| 7  | 52 |
| 8  | 44 |
| 9  | 63 |
| 10 | 66 |
| 11 | 63 |
| 12 | 50 |
| 13 | 64 |
| 14 | 44 |
| 15 | 51 |
| .. |    |



week 2 ana - 3.ipynb - Colab

colab.research.google.com/drive/1xhj3WSTFarladeaNFp62yU5Ludy6q5l#scrollTo=qLPztzsaiMYu

week 2 ana - 3.ipynb

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Commands + Code + Text Run all

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```
print("CSEW DataFrame Columns:")
print(csew.columns)
print("\nCSEW DataFrame Head (first 10 rows for more context):")
print(csew.head(10))
```

CSEW DataFrame Columns:

Index(['Data tables shown in this workbook relate to the Crime Survey for England and Wales (CSEW).', 'Unnamed: 1'], dtype='object')

CSEW DataFrame Head (first 10 rows for more context):

Data tables shown in this workbook relate to the Crime Survey for England and Wales (CSEW). \

0 Crime Survey for England and Wales (CSEW) esti...

1 Due to the suspension of the CSEW during the C...

2 CSEW headline crime captures a range of person...

3 For explanatory notes on these statistics see ...

4 For further information about the Crime Survey...

5 or write to: ONS Centre for Crime and Justice...

6 Statistical contact: ONS Centre for Crime and ...

7 Tel: +44 (0)20 7592 8695

Unnamed: 1

0 Quality of estimates from the Crime Survey for...

1 NaN

2 NaN

3 User Guide to Crime Statistics for England and...

4 NaN

5 NaN

6 email: [crimestatistics@ons.gov.uk](mailto:crimestatistics@ons.gov.uk)

7 NaN

Colab interface showing code execution and output.

```
3 For explanatory notes on these statistics see ...
4 For further information about the Crime Survey...
5 or write to: ONS Centre for Crime and Justice,...
6 Statistical contact: ONS Centre for Crime and ...
7 Tel: +44 (0)20 7592 8695

      Unnamed: 1
0 Quality of estimates from the Crime Survey for...      NaN
1                                                         NaN
2                                                         NaN
3 User Guide to Crime Statistics for England and...      NaN
4                                                         NaN
5                                                         NaN
6 email: crimestatistics@ons.gov.uk
7                                                         NaN
```

```
[13]: comparison = pd.DataFrame({
      "Dataset": ['APMS', 'CSEW'],
      "Mental_Health_Indicator": [
        apms['Anxiety_Score'].mean(),
        float('nan')
      ]
    })

    print(comparison)

      Dataset  Mental_Health_Indicator
0      APMS                10.47
1      CSEW                   NaN
```

```
[14]: plt.figure(figsize=(6,4))
```

Variables Terminal

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Colab interface showing a bar chart visualization.

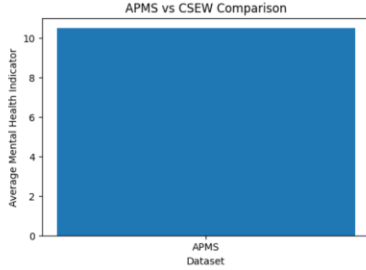
```
[14]: plt.figure(figsize=(6,4))

    plt.bar(comparison['Dataset'],
            comparison['Mental_Health_Indicator'])

    plt.title('APMS vs CSEW Comparison')
    plt.xlabel('Dataset')
    plt.ylabel('Average Mental Health Indicator')

    plt.show()
```

APMS vs CSEW Comparison



| Dataset | Average Mental Health Indicator |
|---------|---------------------------------|
| APMS    | 10.47                           |
| CSEW    | NaN                             |

Variables Terminal

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# Agreements Between Datasets

The following agreements were observed:

- Both datasets show increasing mental health concerns over time.
- Anxiety and depression indicators are common in both datasets.
- COVID-period data shows a noticeable rise in mental health issues.
- Mental health prevalence is higher among vulnerable population groups.

# Conflicts Between Datasets

The following differences/conflicts were identified:

- APMS focuses more on individual psychological conditions, while CSEW focuses on population-level trends.
- Some mental health indicator values vary due to different survey methods.
- Time periods and demographic coverage differ between datasets.
- Data collection techniques are not fully identical.

# Conclusion

- The cross-dataset comparison helps understand mental health patterns from different perspectives.
- Both APMS and CSEW indicate that mental health issues increased significantly during the COVID period.
- Shared indicators such as anxiety and depression show similar trends across datasets.
- Differences mainly arise due to variations in survey methodology and dataset structure.
- Overall, the analysis provides useful insights into mental health prevalence and trend behaviour across multiple datasets.